

VANTARA PETROCHEM - UNIT 3

HEAT EXCHANGER INSPECTION REPORT

Report Number:	INS-2023-E205-001
Equipment Tag:	E-205
Equipment Type:	Shell-and-Tube Heat Exchanger
Inspection Date:	2023-06-20
Inspector:	Arvind Menon (EMP-005)
Inspection Type:	Internal + Tube Bundle Inspection

INSPECTION FINDINGS

Shell Side:

- No significant corrosion observed
- Baffles intact, no erosion at tube holes
- Nozzle welds visually satisfactory

Tube Bundle:

- 3 tubes plugged during this inspection (tubes 45, 78, 112) due to external corrosion pitting
- Estimated 5% tube area loss - acceptable for continued operation
- Fouling deposits observed, chemical cleaning performed

Channel/Tubesheet:

- Tubesheet face in good condition
- Gasket surfaces acceptable
- Channel cover bolting satisfactory

RECOMMENDATIONS

1. Monitor heat transfer performance - efficiency may decrease with plugged tubes
2. Consider tube bundle replacement during next turnaround if plugged tube count exceeds 10%
3. Next inspection due in 24 months or at turnaround

CERTIFICATION

Equipment is FIT FOR SERVICE.

Inspector: Arvind Menon

Date: 2023-06-20